

Week 2 — Six questions for reading a Methods section

Duration: ~2.5 h (Tue, before the Thu lab)

Cap: **150 min**. Markers below total **144**, leaving ~6 minutes of slack.

The timing accounting changed on 2026-09-12. The previous version marked 122 minutes of content and left ~28 minutes unmarked as “the room’s”. That under-filled a 2.5-hour session and hid where the discussion actually goes. The markers below now include the interactive moments — the two vote-and-argue slides and the lab preview block — so the total is what happens in the room. See *If you are running behind* at the foot of this file for the drop order.

The lecture is one argument. The **six questions are the spine**; the NHANES anatomy is what the answers are **made of**, so the component files, variable names and join key are taught in full rather than gestured at. A dedicated preview block before the break puts Thursday’s actual L1 code on screen, so the lab is the execution of Tuesday’s lecture rather than a new topic.

Row-set discipline. Every quantitative slide names the analysis set it is quoting. *Analysis set: defined here, then named on every slide that quotes one.* The reconstruction has two, and they are not interchangeable: **full reconstructed analytic file, N = 6,371** (what scripts 03–06 read) and **locked mortality-linked cohort, N = 6,048** (the WTSAF2YR subset). Saying “N” without saying which is the error this course exists to catch.

Learning objectives

1. Name the **research goal** a paper is pursuing — description, association, prognostic factor, prediction, or causal — and say what evidence that goal would require.
2. Answer the **six questions** for a published study, and show where the paper leaves one of them under-specified.
3. Locate any NHANES quantity a study uses: the **component file**, the **variable name**, and the **join key** — and write the `nhanesA` call that fetches it.
4. Distinguish a **design feature** from **missingness**, and a **row set** from a cohort.
5. Say what a crude-to-adjusted change does and does not establish.

Continuity

- **Last week (Wk1):** reproducibility as a practice, and the course arc.
- **Today:** what a Methods section has to tell you, whether this one does, and what the NHANES files underneath it actually look like.
- **Thursday (L1):** build the merged NHANES frame those answers rest on — previewed here.
- **Toward M0 (Wk3):** ask the same six questions of two or three candidate papers.

Frame · 10 minutes across the slides below

Slide 1 — Where we are in the project (3 min)

- Week 1 argued that a result you cannot rebuild is a claim, not a finding.
- Today is the step before rebuilding: knowing **what** you are trying to rebuild.
- The six questions are the spine; the NHANES anatomy is what the answers are made of.
- Everything named today — component files, variable names, the join key — is used on Thursday in L1 and again for M0.

Slide 2 — Why the question comes before the data (3 min)

- The instinct is to start with “what did they run?”. A model family answers nothing on its own: a Cox model appears under several different research goals.

- Reading a Methods section is the first act of reproduction.
- *MASLD hook*: by the end of today you will have watched this paper answer four of the six questions clearly, one partially, and one not at all.

Slide 3 — The six questions (4 min)

1. What is the **research goal**?
 2. Who is the **population**, and how is eligibility operationalised?
 3. What is the **exposure / index factor**, and the comparison?
 4. What is the **outcome**, time zero, and follow-up?
 5. What **other variables** enter the analysis, and what role are they supposed to play?
 6. What **design and analysis features** must be respected?
- Keep this slide visible: every section below fills one row.
 - **If any one is under-specified, the study is not reconstructable — no matter how clean the results look.**
 - These six are not this paper's; they are the template you take to your own at M0.

Q1 — What is the research goal? · 20 minutes across the slides below

Slide 4 — The paper states two aims, not one (4 min)

- Aim 1: the **clinical correlates** of four groups defined by BMI and waist-to-height ratio.
- Aim 2: the **prognostic value** of that classification for mortality.
- A two-column table contrasts the question and the **time structure**: cross-sectional at baseline versus longitudinal follow-up. One needs the linked mortality file; one does not.

Slide 5 — Five goals the same variables can serve (4 min)

Research goal	Scientific target in this example
Description	What are the characteristics and mortality experience of each group?
Association	Is phenotype statistically associated with subsequent mortality?
Prognostic factor	Does phenotype inform mortality over and above other known factors?
Prediction	How accurately can we predict an individual's risk; does WHtR improve it?
Causal	What would mortality have been under a specified change in central adiposity?

- The variables can be identical. **The target quantity is not.**

Slide 6 — Same variables, different analyses (4 min)

- One table, five rows: typical analysis for each goal, and how other variables are handled under it — stratification, conditional association, other prognostic factors, predictive value with shrinkage, confounder selection by causal knowledge.
- **A Cox model appears in several rows. The model family does not determine the goal.**
- This is the slide that kills “they ran a Cox model, so it is an association study”.

Slide 7 — Same variables, different evidence of success (4 min)

- Description: absolute summaries with uncertainty. Association: HR + CI plus assumptions.
- Prognostic factor: adjusted effect, and for *incremental* value a model with versus without the factor. Prediction: calibration, discrimination, Brier score, validation.
- Causal: a prespecified contrast, positivity/balance diagnostics, sensitivity analyses.
- Do not rush this table — students reuse it at M0 and M1.

Slide 8 — HR + CI: what it is, and is not, evidence for (2 min)

- Enough to report an adjusted association or a prognostic-factor effect.
- **Not** evidence that a prediction model is good, and **not** by itself evidence that a causal effect was identified.
- *MASLD hook*: the paper names “prognostic value” and supplies adjusted association.

Slide 9 — What does this paper most directly support? (2 min)

- Directly supports: description; association; an adjusted prognostic association.
- Does not by itself establish: causation; that reducing WHtR would reduce mortality; that adding WHtR improves individual prediction.
- **Vote and argue**: which row of the ladder is this paper actually on? Take a show of hands before revealing the second half, and do not resolve it too quickly.

Q2 — Who is the population, and how is eligibility operationalised? · 18 minutes for the NHANES anatomy, 14 more after the break

Slide 10 — First: how NHANES is organised (5 min)

- **Six two-year cycles**, 2007–2018, each with a file suffix: E F G H I J. H = 2013–2014, I = 2015–2016, J = 2017–2018 — the letters they will type on Thursday.
- Files are **.XPT** (SAS transport): `haven::read_xpt()`, or `nhanesA::nhanes()`.
- Each file is **one row per participant for one topic** — not one big table.
- Three collection settings: **Interview** (home) · **MEC exam** (measured body size, blood pressure) · **Laboratory** (triglycerides, GGT, glucose).
- Component families: Demographics (DEMO) · Examination (BMX, BPX) · Laboratory (TRIGLY, BIOPRO, GLU, GHB, HDL) · Questionnaire (ALQ, BPQ, DIQ, SMQ).
- **Mortality is not an NHANES file**. It is a separate linked, fixed-width file from NCHS.

Slide 11 — Where the measurements and labs come from (4 min)

Concept	NHANES variable	Component
Participant ID	SEQN	<i>every file — the merge key</i>
Age, sex, race	RIDAGEYR, RIAGENDR, RIDRETH3	DEMO
BMI, waist, height	BMXBMI, BMXWAIST, BMXHT	BMX
Triglycerides	LBXTR	TRIGLY
GGT	LBXSGTSI	BIOPRO
Glucose, HbA1c	LBXGLU, LBXGH	GLU, GHB
HDL cholesterol	LBDHDD	HDL
Blood pressure	BPXSY1-4, BPXDI1-4	BPX

- Do not read the table aloud. Ask instead which rows the **exposure** is built from and which the **eligibility rule** is built from. Both answers are needed in Q3, and the fact that the two lists overlap is the whole of Slide 27.

Slide 12 — Where the questionnaire, design and outcome come from (4 min)

Concept	NHANES variable	Component
BP / lipid medication	BPQ050A, BPQ100D	BPQ
Diabetes	DIQ010, DIQ050, DIQ070	DIQ
Alcohol	ALQ130, ALQ120Q, ALQ120U, ALQ121	ALQ
Survey design	WTMEC2YR, WTSAF2YR, SDMVSTRA, SDMVPSU	DEMO, TRIGLY
Mortality	MORTSTAT, PERMTH_EXM, ELIGSTAT	linked mortality file

- These variables alone reach across **seven examination and laboratory components**, plus demographics, plus questionnaire components, plus a linked mortality file. **That is what L1 assembles.**
- Dwell on the last row: the outcome does not live in NHANES at all. Note too that the fasting weight WTSAF2YR ships on TRIGLY — a design variable living in a lab file.

Slide 13 — SEQN is the join key (5 min)

- Every respondent has one **SEQN**, the identifier CDC uses to sort and merge NHANES files.
- Analysis = pick the topic files you need, then **merge them on SEQN** into one row per person.
- SEQN does **not** repeat across the continuous cycles: all 59,842 values across 2007–2018 are distinct and the per-cycle ranges do not overlap. *Analysis set: all NHANES 2007–2018 records, $N = 59,842$.*
- The ranges are **not contiguous**: a gap of 1,640 sits between 2011–2012 and 2013–2014, so a cycle cannot be recovered arithmetically from an ID.
- Keep a **cycle variable** anyway — provenance, definitions that change between cycles, and weights built per cycle. Not to disambiguate SEQN.

Thursday, previewed: L1 builds exactly this · 14 minutes across the slides below

This block is new. It exists because L1 is not “merge on SEQN” — it is a download, a join, a cycle loop, a stack, a rename, a derivation, a missingness inspection and a descriptive table. Showing the code before the lab turns Thursday into execution rather than a new topic, and every line on these four slides uses a name taught on Slides 10–13.

Slide 14 — L1 in one screen: two files, one key (4 min)

```
pacman::p_load(tidyverse, nhanesA, tableone, naniar)
```

```
demo <- nhanes("DEMO_H", translated = TRUE) # 2013-2014 demographics
bmx <- nhanes("BMX_H", translated = TRUE) # 2013-2014 body measures
```

```
merged <- full_join(demo, bmx, by = "SEQN")
```

- DEMO_H and BMX_H are the same DEMO and BMX from Slide 11, cycle **H**.
- `translated = TRUE` turns stored codes into labels. Convenient — and it changes what a value *looks like*, so recode rules written against raw codes silently miss.
- **full_join keeps people measured in one file but not the other.** A row-set decision made by a single word: `inner_join` would be a different study. Ask which verb they want and why; there is no universally right answer, which is the point.

Slide 15 — Then do it for every cycle (4 min)

```
cycles <- c("H", "I", "J") # 2013-14, 2015-16, 2017-18

all_cycles <- lapply(cycles, function(cy) {
  d <- nhanes(paste0("DEMO_", cy), translated = TRUE)
  b <- nhanes(paste0("BMX_", cy), translated = TRUE)
  full_join(d, b, by = "SEQN") %>% mutate(cycle = cy)
})

stacked <- bind_rows(all_cycles)
```

- `paste0("DEMO_", cy)` is why the suffix table was worth memorising.
- `bind_rows()` stacks cycles; the joins happen **within** a cycle first.
- Our reproduction does the same over all six cycles, E through J.
- *Flag forward to Q5:* a variable present in later cycles may be absent in earlier ones, so a column can appear mid-stack and `bind_rows()` fills the rest with NA. That is a changing instrument, not missingness in the usual sense.

Slide 16 — Raw names in, analysis names out (3 min)

```
analysis <- stacked %>%
  rename(Sex = RIAGENDR, Age = RIDAGEYR,
         RaceEthnicity = RIDRETH3, BMI = BMXBMI) %>%
  filter(Age >= 20) %>%
  mutate(BMICat = cut(BMI, c(0, 18.5, 25, 30, Inf), right = FALSE,
                     labels = c("Underweight", "Normal weight", "Overweight", "Obese")))
```

- Every rename is a **decision you must be able to defend**, not cosmetics.
- **right = FALSE is not optional**, and it is worth thirty seconds. `cut()` defaults to right-closed intervals, which would put BMI exactly 30.0 in *Overweight* — the mirror image of the WHO convention, of the lab's own code, and of reproduction/R/02_build_analytic.R. A boundary convention hidden in a default argument silently reassigns everyone sitting on a cut-point.
- BMICat uses standard WHO cut-points. **The paper does not:** obesity at BMI ≥ 30 , and ≥ 25 for Asian participants.
- The lab keeps **age ≥ 20** ; the paper says **age > 18** ; our reproduction uses **≥ 18** . **Same variable, three eligibility rules, three different studies.** Record which rule you used, every time.

Slide 17 — Look at what is missing before you model it (3 min)

```
colSums(is.na(analysis))
naniar::gg_miss_var(analysis, show_pct = TRUE)

tab1_vars <- c("Age", "RaceEthnicity", "BMI", "BMICat")
CreateTableOne(vars = tab1_vars, data = analysis, strata = "Sex",
               addOverall = TRUE, test = FALSE)
```

- Missingness is inspected first, then decided about — never discovered inside a model.
- `CreateTableOne` is the Table 1 they rebuild for the paper's cohort at M1.
- `test = FALSE` on purpose: a p-value comparing baseline groups answers no question anyone asked.
- Name what Thursday produces: one merged table, a missingness picture, a descriptive table. Week 7 is where the missingness picture becomes a method.

Slide 18 — BREAK (10 min)

Q2, continued — the population this paper defines · 14 minutes across the slides below

Slide 19 — Population: who is being studied? (3 min)

- Adults with **MASLD** in NHANES 2007–2018: excess liver fat with metabolic dysfunction.
- **But nobody looked at a liver.** Steatosis was estimated by a formula.
- The study population is **not the general population** — everyone in the main analysis already meets the study’s own definition.
- The hinge: the population is defined by a computation, so it can be recomputed, and therefore it can disagree.

Slide 20 — How MASLD was actually ascertained (4 min)

- Steatosis estimated with the **Fatty Liver Index** (Bedogni 2006), thresholded at **FLI $\geq$ 60**. A small table on the slide gives its four inputs with their component files: triglycerides (LBXTR, TRIGLY), BMI (BMXBMI, BMX), GGT (LBXSGTSI, BIOPRO), waist (BMXWAIST, BMX). Say explicitly that these enter a published logistic formula and are **not multiplied together** — the slide used to present them --separated, which reads as a product.
- Plus **at least one of five cardiometabolic criteria**: adiposity; raised glucose/HbA1c or diabetes; raised blood pressure or treatment; raised triglycerides or treatment; low HDL.
- So MASLD here is a **derived variable, not a diagnosis** — no imaging, no biopsy.
- Not a criticism: every population study of MASLD in NHANES this decade faces the same constraint. What matters is that eligibility is *computed*, from four named variables in three different component files — two of which return in Q3 as the exposure.

Slide 21 — The funnel, with a type on every row (4 min)

Step	N	Kind of step
NHANES 2007–2018 records	59,842	—
Age $\geq$ 18	36,580	eligibility — drops 23,262
In fasting subsample	14,962	design — drops 21,618
FLI computable	14,170	missingness
Steatosis (FLI $\geq$ 60)	6,052	target population — drops 8,118
MASLD ($\geq$ 1 criterion)	6,050	target population
Mortality-linked	6,048	outcome ascertainment

Analysis set: locked mortality-linked cohort, $N = 6,048$.

- A funnel shows N falling; the **type** column shows why, and the types are not interchangeable.
- **Scope the claim exactly.** The age criterion removes more people than anything else (23,262), but that is the study’s own eligibility rule. **After eligibility**, the two largest drops are **design** (the fasting subsample, 21,618) and the **disease definition** (8,118) — and neither is missing data. The printed drops are on the table so a student can check the sentence rather than take it on trust.
- This scoped form is the course-wide wording: `reproduction/reproduction_report.md`, `Week7_missing_slides.qmd`, the M1 deck and the P1 exemplar all state it this way. An unscoped “the largest single exclusion is design” is false against this table.
- Name the row-set label and say why it is there: the full reconstructed file is 6,371.

Slide 22 — What the funnel reveals (3 min)

- The published paper reports ~6,300 participants with MASLD; the comparable file here is the full reconstructed analytic file, **6,371** (Week 3 works through the difference). The funnel's **6,048** is a further subset, those with a valid fasting weight, which the design-aware analyses use. *Analysis set: full reconstructed analytic file, $N = 6,371$; locked mortality-linked cohort, $N = 6,048$.* The paper's ~6,300 and our 6,371 should not be presented as the same cohort.
 - The paper states eligibility as **age > 18**; this reproduction uses **age ≥ 18** . Small implementation differences are **recorded, not silently harmonised**.
 - Tie back to Slide 16: they have just seen three age rules on one slide. A reproduction that quietly adopts the other rule has destroyed its own evidence.
-

Q3 — What is the exposure, and the comparison? · 16 minutes across the slides below

Slide 23 — The exposure is constructed, not measured (3 min)

- $BMXWAIST \div BMXHT \rightarrow$ **waist-to-height ratio**; $BMXBMI \rightarrow$ **obesity category**; the two categories crossed $\rightarrow$ **four phenotype groups**.
- All three inputs come from **one component, BMX**, in the MEC exam — so a person missing the exam is missing the exposure entirely.
- **Contrast with eligibility**, which needs *BMX and* the laboratory (LBXTR, LBXSGTSI). Eligibility spans two collection settings; the exposure does not.
- *Correction:* an earlier version of this deck said the **exposure** spanned two settings. It does not. Eligibility does.

Slide 24 — Exposure component 1: BMI (4 min)

- Weight and height were **measured** in the exam, not self-reported (BMXBMI).
- BMI marks overall body size — not body fat, and not where fat is stored.
- Obesity defined as **BMI ≥ 30** ; for Asian participants **BMI ≥ 25** (RIDRETH3 == 6).
- **Paper reporting note:** the Methods define obesity as BMI ≥ 30 (≥ 25 for Asian participants), but published Tables 1–2 label the groups as BMI ≥ 25 versus < 25 . Those labels conflict with the Methods *and* with the reported group BMI distributions, so they appear to be a reporting error.
- Ask which version they would implement, and how they would say so in writing. Flag RIDRETH3 — it returns on Slide 32.

Slide 25 — Exposure component 2: waist-to-height ratio (2 min)

- $WHtR = \text{waist (cm)} / \text{height (cm)} = BMXWAIST / BMXHT$. A person 170 cm tall with a 102 cm waist has $WHtR = 0.60$ — worked on screen so the threshold stops being abstract.
- High central adiposity defined as **WHtR ≥ 0.60** ; a simple marker of abdominal adiposity, not a CT or MRI measurement of visceral fat.

Slide 26 — The exposure is the combination (4 min)

	WHtR < 0.60	WHtR ≥ 0.60
Obesity by BMI	Group I — <i>reference</i>	Group II
No obesity by BMI	Group III	Group IV

- **Group IV** — not obese by BMI, large waist for height. The study asks whether BMI alone misses this higher-risk phenotype.

- There is no treated arm and control arm: the four-category phenotype is the **index factor**, and mortality is compared **across** the categories.
- **Group I is the reference.** A ratio without its referent means nothing — and here the referent is an **obese** group. Make them say the Group IV versus Group I contrast aloud.

Slide 27 — Eligibility and exposure share their inputs (3 min)

Inputs	
Eligibility ($FLI \geq 60$)	triglycerides, GGT, BMI , waist
Exposure (phenotype)	BMI , waist $\div$ height

- Being selected into the study and being placed in a phenotype group are **not independent events**. That is **structural dependence**.
- For a causal question, conditioning on this selected population could open selection or collider paths under some structures; for a prognostic question, it primarily defines a selected target population. Not automatically circularity, not automatically bias.
- We name it today; **M1** dissects it. Takeaway: **never trust a label you have not recomputed**.

Q4 — What is the outcome, time zero, and follow-up? · 11 minutes across the slides below

Slide 28 — Outcome and the clock (6 min)

- Outcome: **all-cause mortality** (MORTSTAT), via the NCHS linked mortality file — not an NHANES file, a separate linked, fixed-width one.
- **Time zero is the MEC examination.** PERMTH_EXM is person-months from that exam.
- Follow-up runs through **31 December 2019**, stated in the NCHS 2019 Linked Mortality File linkage methodology document — not inferred from the filename. That sourcing is the standard we ask of them.
- Participants entered between 2007 and 2018, so **follow-up lengths and censoring differ**. Time-to-event methods are appropriate — and a **risk difference at a fixed horizon can still be estimated** with methods that account for censoring.
- *Correction:* an earlier version said differing follow-up meant “survival analysis, not a risk difference”. Differing follow-up forbids a *naive* absolute contrast, not an absolute contrast.

Slide 29 — Time zero is a question the checklist never asks (5 min)

- “Exposure, outcome, covariates” can all be specified while time zero stays ambiguous, and an ambiguous time zero silently changes who is at risk and for how long.
- *MASLD hook:* exposure here is measured **at** time zero, which is convenient and rare.
- Ask of your M0 candidates: when does follow-up begin, and does everyone start together? A show of hands on whether their candidates state it is worth the thirty seconds.

Q5 — What other variables enter, and in what role? · 15 minutes across the slides below

Slide 30 — In the model is not the same as confounding (4 min)

- The paper adjusts for age, sex, and a list of comorbidities and medications.
- “We adjusted for X” says where X sits in the syntax. It does not say what X **does**.

- The same variable can be a confounder, a mediator, a competing prognostic factor, or a collider — and which it is depends on the research goal from Q1. Under a prognostic-factor question the adjustment set need not be confounders at all.

Slide 31 — What a crude-to-adjusted change establishes (5 min)

- Group IV vs Group I, all-cause HR ≈ 15.1 crude $\rightarrow \approx 2.9$ with the paper's Model 2 covariates, both unweighted, as in the paper. *Analysis set: full reconstructed analytic file, $N = 6,371$; Model 2 drops rows missing one of its covariates, so the rows change slightly too.*
- This is the paper replication: the paper's contrast and its model. Week 4 makes the same kind of move on other contrasts and adjustment sets, so its numbers differ.
- Conditioning on the adjustment set moves the estimate enormously. What that movement **means** — confounding controlled, mediation removed, a collider opened — depends on the goal and on the causal role of those variables.
- **Week 4 does that work.** Today it is enough to see that the size of a move is not itself an interpretation.
- **Vote and argue:** what would you need to know to call this confounding? Collect answers; resist supplying one.

Slide 32 — Five NHANES traps, each of which has bitten this project (3 min)

- **ALQ130 is drinks per *drinking day*, not drinks per day.** The paper's ">2 drinks a day" is an average intake — frequency $\times$ quantity. Applied to ALQ130 directly, the rule removes **2,063** people here; the average-intake rule removes **427**, all of whom the direct rule also removes, so the direct rule over-excludes about **1,600** (1,636). *Analysis set: full reconstructed analytic file, $N = 6,371$.* **Week 3 owns the full demonstration.**
- **777 = refused, 999 = don't know.** Codes, not counts. Left inside a threshold rule they reclassified **nine** records (refused or don't know) as heavy drinkers (the exclusion moves $2,063 \rightarrow 2,072$).
- Keep those two bullets **separate**. An earlier version of this slide fused them and attached the direct rule's ~2,000 removals to the reserved codes — wrong by a factor of about two hundred, and the sort of error this course exists to catch.
- **RIDRETH3 does not exist before 2011.** It identifies Asian participants for the lower BMI threshold, so the first two cycles cannot apply the paper's own rule. A definition that cannot be applied in a third of your cycles is a finding, not a nuisance.
- **Fasting labs carry their own weight:** LBXTR and LBXGLU come from the fasting subsample, weighted by WTSF2YR, not WTMEC2YR.
- **Blood pressure is not a row mean** — next slide.
- These sit under Q5 rather than in a separate tips section because each is a variable-handling decision.

Slide 33 — The blood pressure trap, in full (3 min)

- The NHANES **Blood Pressure Procedures Manual (s3.1.7.3)** specifies the averaging: the **first reading is excluded** whenever more than one was obtained; an **all-zero diastolic set averages to zero**, because a diastolic 0 is a measurement and a failure to obtain a reading is coded separately as missing; and a **zero beside a positive reading is omitted**.
- Applying it changed the derived diastolic pressure for **5,523 of 6,371** participants and moved hypertension from 3,444 to 3,448, while the Model 1 and Model 2 hazard ratios of the reproduced Table 2 shifted only in the third significant figure. *Analysis set: full reconstructed analytic file, $N = 6,371$.*
- **A coding error can be real, and worth fixing, even when the headline estimate survives.**
- The paper does not specify its own averaging in enough detail to establish that it used this algorithm, so ours is a documentation-informed choice, not a claim of exact computational reproduction.

Q6 — What design and analysis features must be respected? · 10 minutes across the slides below

Slide 34 — NHANES is not a simple random sample (3 min)

- A complex multistage probability design: **weights, strata (SDMVSTRA), PSUs (SDMVPSU)**. We *use* them in Week 6; today, note they exist and that ignoring them is a choice with consequences.
- Fasting-subsample analyses carry their own weight, **WTSAF2YR**, not WTMEC2YR.
- Build the design on the **full frame**, then `subset()` — never on the subsample alone. The single most common survey-analysis error, worth planting now.

Slide 35 — Designed subsampling is not missingness (4 min)

- Sampling into the fasting subsample is **statistically designed**, and it carries a weight built to keep it nationally representative. It is not purely design, though: **2,031** adults carry a fasting weight of exactly 0, and fasting compliance and laboratory nonresponse can also affect who has a positive weight (Week 3 says the same).
- Check it rather than assume: adults in and out of it differ by **0.2 years of age, 0.02 kg/m² of BMI and 0.07 cm of waist**. It is balanced on the very measurements this paper's exposure is built from. *Analysis set: all NHANES adults 2007–2018, N = 36,580.*
- So the distinction is **designed subsampling** $\neq$ **ordinary missingness**. That it was *checked*, rather than assumed in either direction, is the transferable habit.

Slide 36 — Survey design: paper versus reproduction (3 min)

- The published paper's Statistical analysis section **does not report** using NHANES survey weights, strata or PSUs. Our pipeline does include them.
- Therefore a survey-weighted rebuild is a **design-informed reanalysis**, not a literal reproduction of every published analytic choice.
- Without the design information, estimates and standard errors should not be described as design-based nationally representative estimates.
- *MASLD hook*: this is Q6 answered by omission — a documented divergence that is neither our error nor theirs, and the honest move is to label it as one.

Closing · 6 minutes across the slides below

Slide 37 — PECOT, and the question rebuilt (2 min)

Element	This study
P — Population	Adults with MASLD by $FLI \geq 60 + \geq 1$ cardiometabolic criterion, in the fasting subsample
E — Exposure	Four-category BMI $\times$ WHtR phenotype
C — Comparison	Groups II, III, IV versus Group I
O — Outcome	All-cause mortality to 31 Dec 2019
T — Time	Months from exam to death or censoring

- *Among adults with MASLD, are BMI/WHtR phenotypes associated with different subsequent risks of all-cause mortality?* That sentence is the deliverable shape for M0.

Slide 38 — M0: ask the six of your own papers (2 min)

- Shortlist two or three candidate NHANES papers and answer all six questions for each, each answer with a pointer to where in the paper you found it.

- **Reconstructable is necessary, not sufficient.** A paper can be perfectly specified and still be a bad M0 choice — too few events, no design feature worth respecting, no variable whose handling you could interrogate.
- Where a paper leaves a question unanswered, that is not a disqualification — it is a candidate for the judgement you will defend at M1.
- If you are ahead of schedule, work one candidate live. It is the best use of spare minutes.

Slide 39 — Bridge to L1 and M0 (2 min)

- **Thursday (L1):** the code from before the break — several cycles downloaded, topic files read, merged on SEQN, renamed, categorised, missingness inspected, Table 1 built.
 - **Before you go:** you should be able to name the component file each variable on these slides comes from. Ask two or three at random on the way out.
 - **Toward M0 (Wk3):** bring six answers for each candidate paper, not a summary.
 - Takeaway: reading a Methods section *is* the first act of reproduction.
-

If you are running behind

Drop in this order. Each is a slide whose content is carried elsewhere in the course, so cutting it costs continuity rather than a concept:

1. **Slide 7** (different evidence of success, 4 min) — Slide 8 states its one-line version.
2. **Slide 25** (WHtR, 2 min) — the ratio is defined again on Slide 26's table header.
3. **Slide 6** (different analyses, 4 min) — painful, but Slide 5 carries the distinction.
4. **Slide 33** (the blood pressure trap, 3 min) — Slide 32's blood-pressure bullet survives alone. (Named, not numbered: the bullet's position moved once already.)
5. **Slide 22** (what the funnel reveals, 3 min) — Week 3 revisits the cohort arithmetic.

Do **not** cut Slides 14–17. The lab preview is the reason this deck was rebuilt, and dropping it puts students into L1 cold.

Deliberately deferred

Three things the old Week 2 taught are **not** here, and are deferred on purpose rather than squeezed under a question:

- **The alcohol-exclusion three-way demonstration** (~23 min including live work). It is a cohort-construction exercise and Week 3 owns it (`Week3_cohort_construction_FLI.md`). Its residue survives as the **two alcohol bullets** on Slide 32 — the ALQ130 drinks-per-drinking-day trap and the 777/999 reserved-code trap — which name the two errors and their sizes without running the demonstration.
- **“Table 1 reproduces closely, therefore the variable definitions are recoverable.”** The inference is exactly what Q5 warns against, and Week 4 carries the careful version.
- **The four primitives** (exposure / outcome / eligibility / variables). Kept as an idea, demoted from *the* framework to a reconstructability check nested inside Q2–Q5. It never asked what kind of question the variables were meant to answer, which is why a crude-to-adjusted collapse could be read as settled confounding.

The **reading-Methods traps checklist** is no longer listed as deferred: the NHANES traps now have their own slides under Q5, and the six questions are the reading checklist.

If a later revision wants any of the three back, they belong where they are now owned, not retrofitted into a six-question spine.

Relationship to the companion deck

lectures/bmj_open_revealjs/ — *Decomposing the Study Aim* — is the standalone deck this rebuild draws its NHANES anatomy, PECOT and goal-ladder material from. It remains published as a companion for students who want the aim-decomposition argument on its own, without the six-question scaffolding or the lab preview. **Where the two disagree, this file wins.**

Two errors were corrected during this rebuild. **Neither is in the companion today**, but they reached Week 2 by different routes, and the difference matters — an inherited error attributed to the wrong source cannot be traced:

- **The exposure spans one collection setting, not two** (Slide 23). This one was never in the companion, which makes no collection-settings claim about the exposure at all. It was introduced by the previous Week 2 rewrite (c782167).
- **Differing follow-up permits an absolute contrast** computed with censoring in mind, and forbids only a naive one (Slide 28). This one **was inherited from the companion**, which carried “survival analysis and not a risk difference” in its first version (02e50be) until ae5e3ef corrected it. The previous Week 2 rebuild copied the *pre-correction* wording an hour and a half later — a fix that lands in one file and not in its copy is exactly how a corrected error comes back.

One claim **was** wrong in the companion and has been fixed there in the same change: its funnel slide called the fasting subsample “the largest single exclusion”, which is false against its own table. It now carries the course-canonical *after eligibility* scoping, and its fused ALQ130 / 777-999 trap bullet has been split the same way as Slide 32’s.